1. By what percentage will the transmission range of a TV tower be affected when the height of the tower is increased by 21%?
[2023 (06 Apr Shift 1)]
(1) 15%
(2) 12%
(3) 10%
(4) 14%

2. For an amplitude modulated wave the minimum amplitude is 3 V, while the modulation index is 60%. The maximum amplitude of the modulated wave is:
[2023 (06 Apr Shift 2)]
(1) 5 V
(2) 15 V
(3) 12 V
(4) 10 V

3. A TV transmitting antenna is 98 m high and the receiving antenna is at the ground level. If the radius of the earth is 6400 km, the surface area covered by the transmitting antenna is approximately:
[2023 (08 Apr Shift 1)]
(1) 1240 km^2
(2) 3942 km^2
(3) 4868 km^2
(4) 1549 km^2

4. The power radiated from a linear antenna of length l is proportional to
(Given, $\lambda =$ Wavelength of wave):
[2023 (08 Apr Shift 2)]
(1) $\frac{l}{\lambda}$
(2) $\left(\frac{l}{\lambda}\right)^2$
(3) $\frac{l}{\lambda^2}$
(4) $\frac{l^2}{\lambda}$

5. A carrier wave of amplitude 15 V is modulated by a sinusoidal base band signal of amplitude 3 V. The ratio of maximum amplitude to minimum amplitude in an amplitude modulated wave is
[2023 (10 Apr Shift 1)]
(1) 2
(2) $\frac{3}{2}$
(3) 1
(4) 5

6. A massage signal of frequency 3 kHz is used to modulate a carrier signal of frequency 1.5 MHz. The bandwidth of the amplitude modulated wave is
[2023 (10 Apr Shift 2)]
(1) 6 MHz
(2) 6 kHz
(3) 3 MHz
(4) 3 kHz

7. A transmitting antenna is kept on the surface of the earth. The minimum height of receiving antenna required to receive the signal in line of sight at 4 km distance from it is $x \times 10^{-2}$ m. The value of x is _____.
(Let, radius of earth $R = 6400 \text{ km}$)
[2023 (11 Apr Shift 1)]
(1) 125
(2) 1250
(3) 12.5
(4) 1.25

8. In satellite communication, the uplink frequency band used is:
[2023 (11 Apr Shift 2)]
(1) $420 - 890 \text{ MHz}$
(2) $5.925 - 6.425 \text{ GHz}$
(3) $76 - 88 \text{ MHz}$
(4) $3.7 - 4.2 \text{ GHz}$

9. The amplitude of $15 \sin(1000 \pi t)$ is modulated by $10 \sin(4 \pi t)$ signal. The amplitude modulated signal contains frequencies of

A. 500 Hz

B. 2 Hz

C. 250 Hz

D. 498 Hz

E. 502 Hz

Choose the correct answer from the options given below

[2023 (12 Apr Shift 1)]

(1) A and B only

(2) A and C only

(3) A and D only

(4) A, D and E only

10. Match List-I with List-II

	List-I (Layer of atmosphere)		List-II (Approximate height over earth's surface)
(A)	F_1−Layer	(I)	10 km
(B)	D−Layer	(II)	$170-190 \text{ km}$
(C)	Troposphere	(III)	100 km
(D)	E−Layer	(V)	$65-75 \text{ km}$

Choose the correct answer from the options given below.

[2023 (13 Apr Shift 1)]

(1) A− II, B− IV, C− I, D− III

(2) A− II, B− IV, C− III, D− I

(3) A− III, B− IV, C− I, D− II

(4) A− II, B− I, C− IV, D− III

11. To radiate EM signal of wavelength λ with high efficiency, the antennas should have a minimum size equal to:

[2023 (13 Apr Shift 2)]

(1) 2λ

(2) $\frac{\lambda}{2}$

(3) $\frac{\lambda}{4}$

(4) λ

12. The height of transmitting antenna is 180 m and the height of the receiving antenna is 245 m. The maximum distance between them for satisfactory communication in line of sight will be: (given $R = 6400 \text{ km}$)

[2023 (15 Apr Shift 1)]

(1) 96 km

(2) 56 km

(3) 48 km

(4) 104 km

ANSWER KEYS

1. (3)	**2.** (3)	**3.** (2)	**4.** (2)	**5.** (2)	**6.** (2)	**7.** (1)	**8.** (2)
9. (4)	**10.** (1)	**11.** (3)	**12.** (4)				

1. (3)

The range of a TV tower antenna is given by

$d=\sqrt{2hR}$

New height is given as

$h' = 1.21h$

Let the new range be

$d' = \sqrt{2h'R}$

$\Rightarrow d' = \sqrt{1.21}\sqrt{2hR} = 1.1d$

So, the increase in the range is $= \frac{1.1d-d}{d}\times 100 = 10\%$

2. (3)

Modulatio index is the ratio of the difference in amplitudes to the sum of the amplitudes.

The formula to calculate the modulation index of a wave is given by

$\mu = \frac{A_{\max}-A_{\min}}{A_{\max}+A_{\min}} \quad \ldots(1)$

Substitute the values of the known parameters into equation (1) and solve to calculate the required value of maximum amplitude.

$0.6 = \frac{A_{\max}-3}{A_{\max}+3}$

$\Rightarrow 0.6A_{\max}+1.8 = A_{\max}-3$

$\Rightarrow 0.4A_{\max} = 4.8$

$\Rightarrow A_{\max} = \frac{4.8}{0.4}$

$= 12\text{ V}$

3. (2)

The given data is

$H = 98\text{ m}$

$R = 6400\text{ km}$

The line of sight formula for an antenna is

$d=\sqrt{2RH}$

$\Rightarrow \text{Area} = \pi d^2 = \pi\times 2\times R\times h = 2\times 6400\times 98\times 10^{-3}\times\pi = 3940\text{ km}^2$

4. (2)

Linear antennas are considered to be those antennas that imply the use of electrically thin conductors having diameter much less than the wavelength of the electromagnetic wave passing through them.

The formula to calculate the effective power radiated by a linear antenna is given by

$P_{eff} = k\frac{l^2}{\lambda^2}$

where, k is a constant.

5. (2)

The given data is

$A_c = 15\text{ V}$

$A_m = 3\text{ V}$

Thus, the maximum amplitude and minimum amplitude is

$A_{max} = A_c + A_m$

$A_{min} = A_c - A_m$

Hence, the ratio is

$\frac{A_{\max}}{A_{\min}} = \frac{15+3}{15-3} = \frac{18}{12} = \left(\frac{3}{2}\right)$

6. (2)

The relation between bandwidth and the signal frequency is given by

$\text{Bandwidth} = 2 \times \text{ signal frequency} \quad \ldots(1)$

Substitute the values of the known parameter into equation (1) to calculate the required bandwidth.

$\text{Bandwidth} = 2 \times 3 \text{ kHz}$

$= 6 \text{ kHz}$

7. (1)

The formula to calculate the range of the antenna is given by

$\gamma = \sqrt{2Rh} \quad \ldots(1)$

Substitute the values of the known parameters into equation (1) and solve to obtain the required height of the antenna.

$4 \times 10^3 = \sqrt{2 \times 6400000 \times h}$

$\Rightarrow 16 \times 10^6 = 2 \times 64 \times 10^5 \times h$

$\Rightarrow h = \left(\frac{160}{2 \times 64}\right) \text{m} = \left(\frac{10}{8}\right) \text{m}$

$= \frac{1000}{8} \times 10^{-2} \text{ m}$

$= 125 \times 10^{-2} \text{ m}$

8. (2)

In satellite telecommunication, a downlink is the link from a satellite down to one or more ground stations or receivers, and an uplink is the link from a ground station up to a satellite. Uplink frequency band for satellite communication is $5.925 - 6.425 \text{ GHz}$.

9. (4)

The carrier wave frequency is given by

$f_c = \frac{1000\pi}{2\pi}$

$= 500 \text{ Hz}$

The modulating wave frequency is given by

$f_m = \frac{4\pi}{2\pi}$

$= 2 \text{ Hz}$

Hence, the frequencies contained in the modulated signal are given by

$f = f_c - f_m, f_m, f_c + f_m$

$= 498 \text{ Hz}, 500 \text{ Hz}, 502 \text{ Hz}$

10. (1)

The F_1 layer is above the E layer and the ionization is less compared to E layer. It ranges to $170 - 190 \text{ km}$.

The D layer is located above the stratosphere. It ranges from $60 - 90 \text{ km}$ above the earth's surface.

The troposphere starts from the ground and ranges to 100 km.

The E layer is located above the D layer. The intensity of ionization is less than the D layer. It ranges from $100 - 125 \text{ km}$ above the earth's surface.

11. (3)

For efficient radiation and reception, the height of transmitting and receiving antennas should be comparable to a quarter of wavelength of the frequency used. So for efficient transmission of signals of wavelengths λ the minimum length of antenna should be $\frac{\lambda}{4}$.

12. (4)

The maximum distance for line of sight communication is

$d = \sqrt{2hR} + \sqrt{2h'R}$

The given data is

$R = 6400 \text{ km}$

$h = 180 \text{ m}$

$h' = 245 \text{ m}$

Thus, the value is

$d = \sqrt{2 \times 180 \times 6400 \times 10^3}$

$+ \sqrt{2 \times 245 \times 6400 \times 10^3}$

$= 6 \times 80 \times 10^2 + 7 \times 80 \times 10^2$

$= 104 \text{ km}$